

Cloud Computing

Presented by: Dr M Chere



CLOUD COMPUTING:
Master the Concepts, Architecture, and Applications with Real-world examples and Case studies

Kamal Kant Hiran Ruchi Doshi Dr. Temitayo Fagbola Mehul Mahrish

Lesson 3: Cloud Computing Architecture

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"Architecture is a **visual art** and the buildings speak for themselves."

Julia Morgan, American architect

Objectives

- To discuss the cloud computing architecture design principles
- To discuss the cloud computing life cycle (CCLC) phases in detail
- To learn about the cloud computing reference architecture
- To learn about the concepts of load balancing approach
- To learn about mobile cloud computing (MCC)
- To discuss the Oracle cloud management as a case study

Content



Cloud Computing
Architecture
Design Principles



Cloud Computing
Life Cycle



Cloud Computing
Reference
Architecture



Load Balancing
Approach



Mobile Cloud
Computing (MCC)

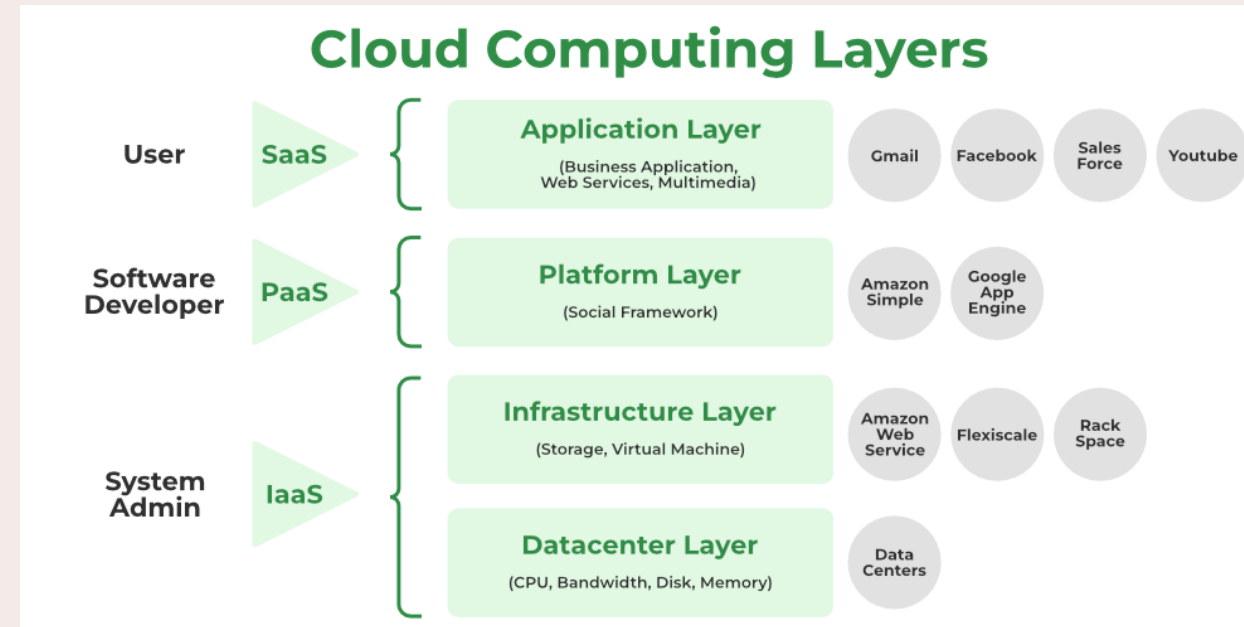
(Case study - Oracle Cloud Management)

Cloud Computing Architecture

- Defined as the **structured design** of **systems** that **deliver computer services over the internet** comprising components such as **servers, storage, and networks**, to **provide scalable and flexible computing resources** over the internet

Cloud Computing Architecture

- Key Categories:
 - Front-end:
 - User interfaces and client devices such as web applications, browsers, mobile devices
 - Back-end: the cloud, which provides data storage, dedicated servers, operating systems, and security measures



<https://www.geeksforgeeks.org/layered-architecture-of-cloud/>

Cloud Computing Architecture

Some Components

Infrastructure

Hardware & Software:

- Servers
- Storage
- Network devices
- Software for managing & operating cloud services

Management Software

Middleware:

- Coordinates communication between the **front-end** and **back-end**,
- Ensuring smooth **user experiences** and **real-time service delivery**

Security

Security Tools:

- Secure cloud resources, systems, files, and infrastructure to end-users
- **Protect against cyberattacks** and **system failures**, employing measures like **virtual firewalls**, **encryption**, **access restrictions**, and **authentication protocols**

Seven Cloud Computing Architecture Design Principles

Cloud Computing Design Principles	Description
Adaptive and elastic	Cloud architecture should foster growth of users, traffic, or data size with no drop in operational performance
Implement loose coupling	Cloud architecture should ideally be designed in such a way that software and hardware components are made separate and distinct to reduce interdependencies
Managed database services	Cloud architecture should help remove constraints that arise with licensing costs and the ability to support diverse database engines that were an inherent part of the traditional IT infrastructure

Seven Cloud Computing Architecture Design Principles

Cloud Computing Design Principle	Description
Be sure to remove single points of failure	A cloud computing system is consistently accessible if it can endure the failure of a single component or multiple ones e.g., hard disks, servers, network links, operating systems, etc.)
Optimize cost	Cloud architecture should be designed to optimize costs , considering that costs can be reduced by selecting the configurations and data storage arrangements to suit users' needs by taking advantage of a variety of Instance Purchasing choices
Caching	Data caching should be applied to multiple layers of the cloud architecture for optimum application performance and efficiency of costs

Seven Cloud Computing Architecture Design Principles

Cloud Computing Design Principle	Description
Security	Both organizations and individual users expect their private information to not only be safe but also secure . Therefore, cloud architecture should be designed notwithstanding legal obligations for providers to adhere to high data security standards

Note: Cloud service providers' claims of **dependability, accessibility, security, and performance** must be certified by some IT authority

Benefits of Cloud Computing Architecture Design Principles

Benefits	Description
Transparent architecture and control	Cloud service consumers should have access to the design and operation of the cloud system
Improved productivity	Cloud systems should provide significant improvements in the levels of efficiency and productivity than those in traditional IT settings
Assured data protection	Cloud service consumers should be confident that data privacy standards and regulations are being followed by the provider
Automate operations	Cloud service consumers' platform services runtime should have manual operations in marginal amounts Accessibility should not be hindered by expected hardware failures
Robust identity domain separation	Cloud service consumers should not be subjected to any impairment caused by other users

Cloud Computing Life Cycle (CCLC)

Cloud Computing Life Cycle (CCLC)

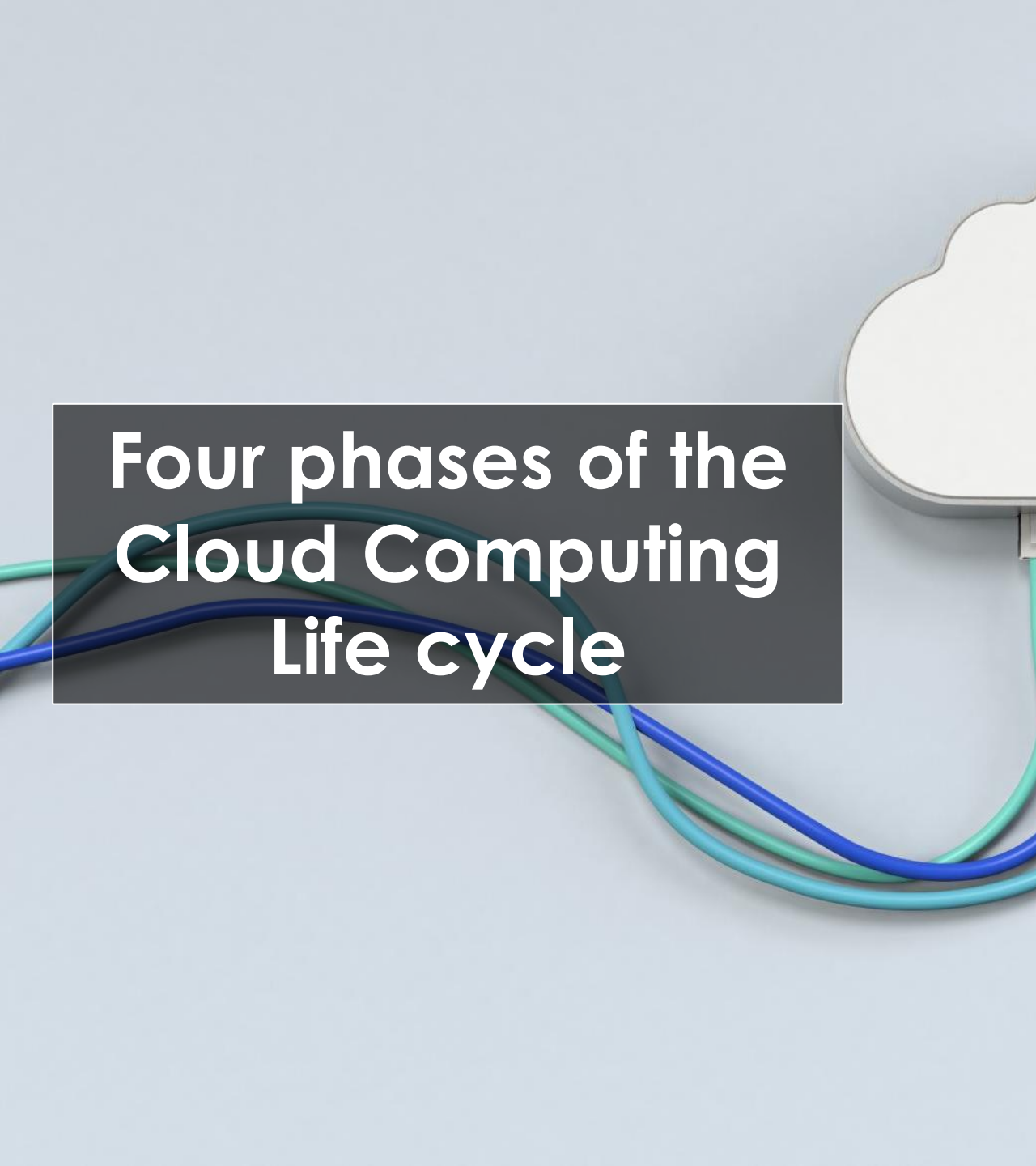
- Developed by a group of organizations using an open innovation approach encompassing **nine (9) steps** that can be followed for both the **migration to cloud computing** and **continuous management of public cloud-based services**
- Each step in this life cycle is described in terms of the **primary challenges faced**, and the **prescribed activities**, with **ensuing outputs required to resolve these problems**

Objectives of Cloud Computing Life Cycle (CCLC)

- Remembering cloud computing has the potential of noteworthy benefits, **including** but **not limited** to *reduced costs, better service delivery*, and a move to a **pay-per-use model**
- **Objective of the CCLC were formulated to:**
 - Negotiate the ever-changing nature of the cloud environment
 - Accelerate service speed
 - Increase flexibility
 - Satisfy the needs of individuals and organizations

Objectives of Cloud Computing Life Cycle (CCLC)

- **Four stages** within the cloud life cycle
 - Segmented into **nine steps**
 - Sequential order



Four phases of the Cloud Computing Life cycle

- A – Architect
- E – Engage
- O – Operate
- R – Refresh

Four phases of the Cloud Computing Life Cycle

Architect

- This phase begins with the **analysis** and **preparation** of the cloud scheme
- Organization will only commit a minimal number of resources to decide if they should go forward with comprehensive development

Engage

- **Selecting** a **cloud service provider** that can provide the necessary services
- Most **organizations** **decide to stop** at this stage when the **suitable cloud services** are **not offered**, or because **they do not have the assurance** that any cloud provider can deliver the required services

Four phases of the Cloud Computing Life Cycle

Operate

- Comprises the **execution of** and **regular management** of the cloud service

Reflect

- Includes the **continued appraisal** of cloud services provided

Four Phases of the Cloud Computing Life Cycle

Architect

Investigate

Identify

Implementation strategy

Business design

Engage

Select

Negotiate

Operate

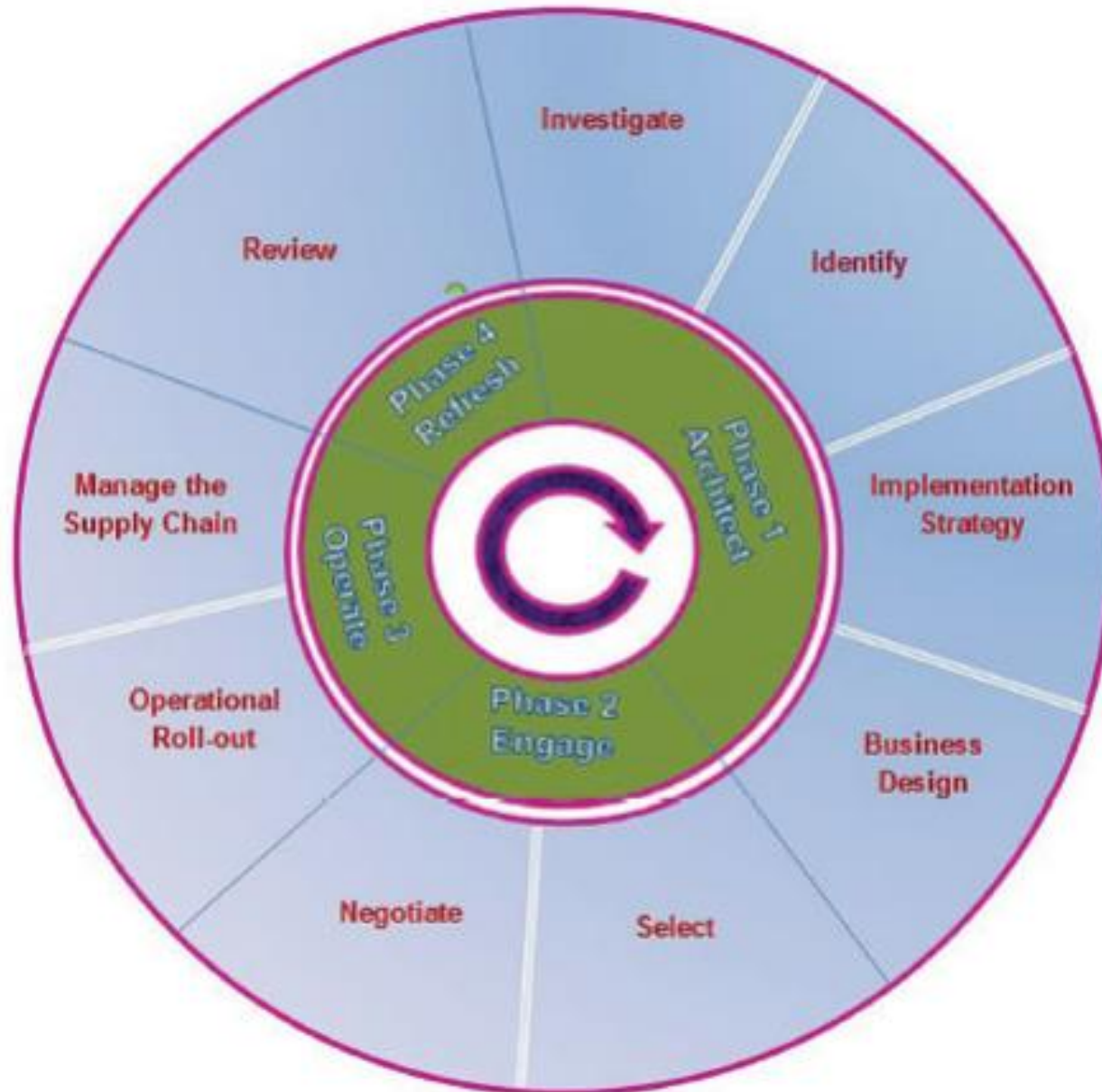
Operational roll-out

Manage the chain

Refresh

Review

Cloud Computing Life Cycle



Phase 1 - Architect

Step 1: Investigate	Step 2: Identify	Step 3: Strategy Implementation	Step 4: Business Design
▪ Examine prospects ▪ Provide an understanding of what firms want to achieve by moving to the cloud, and how they expect to accomplish specific objectives and overall goals	▪ Classify facilities ▪ Impartially evaluate the components of business suitable to outsource to the cloud and the effect it will have on the firm's current delivery model	▪ Define services ▪ Outline a strategy of how the outsourced cloud services will be implemented ▪ During this step, key decisions on employment, communication, project execution, organizational policies, and risk assessment strategies are well-defined	▪ Design model ▪ Create a detailed service model of the proposed outsourced cloud service package ▪ This plan explains the new service, its operation management, how it fits into the existing business systems, and how results will be reported

Phase 2 - Engage

Step 5: Select

Select supplier - choose the best cloud provider based on:

- Interface
- Quality
- Cost, and continuity

Step 6: Negotiate

- Negotiate service terms
- Finalize discussions based on **design criteria** and **sign service contracts**

Phase 3 - Operate

Step 7: Operational Roll-out

- Implement plan
- **Assemble** a team that will **manage the migration** to the new cloud platform
- **Ensures** communication is disseminated to all stakeholders via email, letters, faxes, bulletins, and so on regarding the transition of the new services provided

Step 8: Manage the Chain

- Manage the supply chain
- Manage new cloud services
- The company staff will need to familiarize themselves with the new cloud interface, particularly at the IT management level

Step: 9: Review

- Periodically review cloud service requirements based on the service itself,
- Other changes within the business,
- Changes within the supplier organization, or
- The need to change the supplier

Note:

Changing the supplier can create some difficulties, so carefully consider

Step 5: Select

Rationale for using Life Cycle stages

Pp. 148 - 150

Cloud Computing Reference Architecture

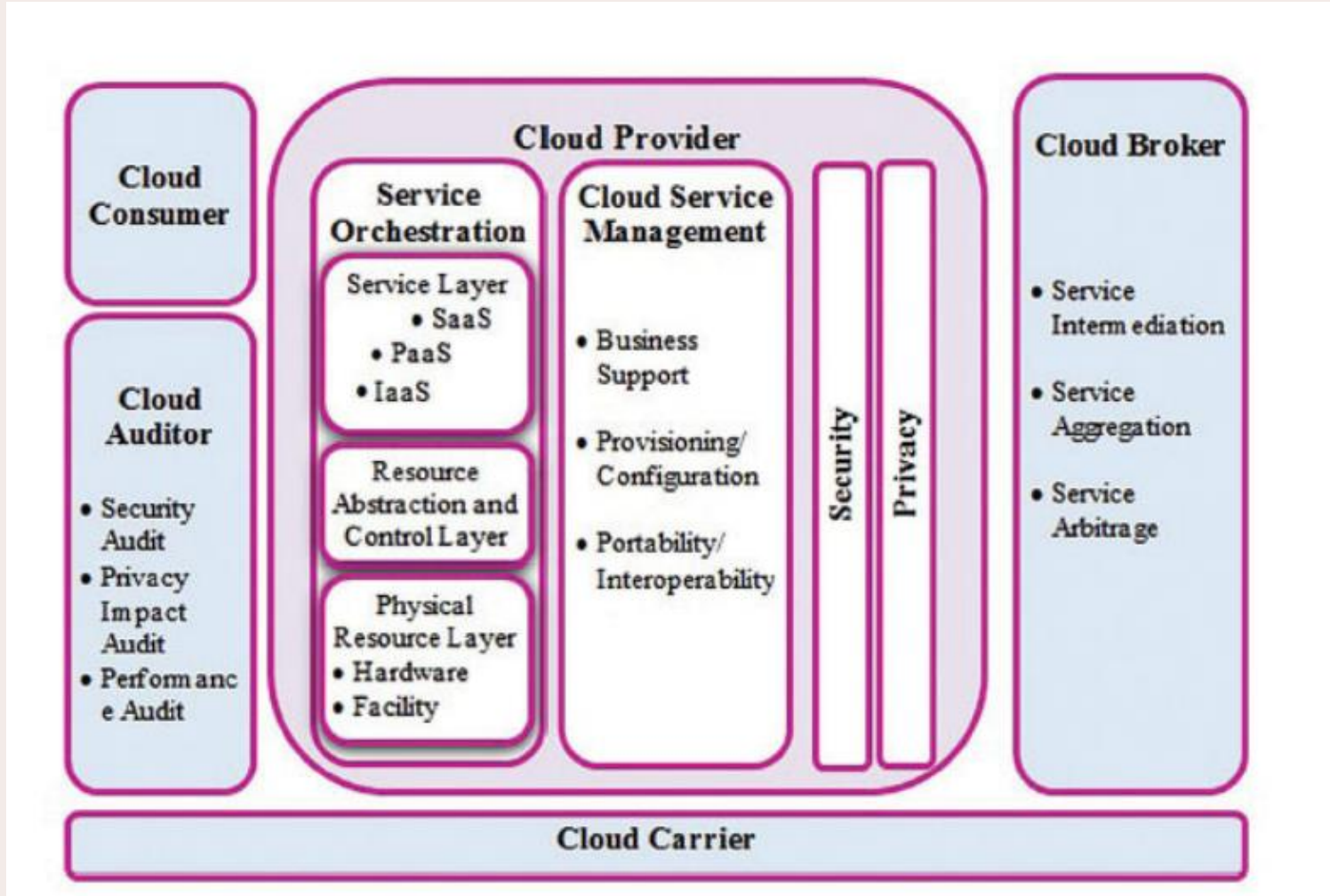
Cloud Computing Reference Architecture

- Refers to a **practical plan** for a system with a distinct range of operations to:
 - Ensure **quality** and **consistency** over the course of system development and **delivery projects**

The Conceptual Reference Model

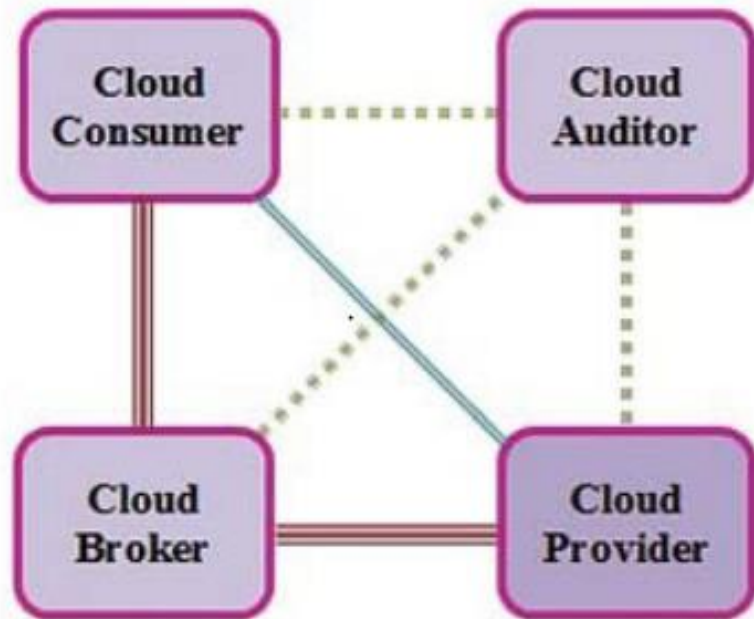
- **National Institute of Standards and Technology (NIST)** cloud computing reference architecture:
 - Shows **five (5) main participants** in the cloud computing environment and the **roles they play**
 - Cloud Consumer
 - Cloud Provider
 - Cloud Carrier
 - Cloud Auditor
 - Cloud Broker

NIST Cloud Computing Reference Architecture



Each participant is either an **individual** or an **entity that participates** in a transaction or **process** and/or performs various tasks within the cloud system

Interactions between the Actors in Cloud Computing



- The communication path between a Cloud Provider and a Cloud Consumer
- - The communication path for a Cloud Auditor to cloud auditing information
- ≡ The communication paths for a Cloud Broker to provide service to a Cloud Consumer

Cloud consumers may request services **directly** from **providers** or through **brokers**, while **Auditors** may collect relevant information from any of the **three aforementioned parties** in efforts to conduct independent audits

Primary Stakeholders in Cloud Computing Environment

Cloud Consumer	Cloud Provider	Cloud Carrier	Cloud Auditor	Cloud Broker
- The primary stakeholder for the cloud computing service - Cloud consumer is the end user who browses or uses the services offered by Cloud Service Providers (CSP) and enters into service agreements with the cloud provider - The cloud user makes payments each time they use a service - billed on a pay-per-use basis	- Any individual, group, or other entity in charge of making a service accessible to cloud users - Creates the requested software, platforms, and infrastructure services, - Manages the technical infrastructure needed to supply the services, - Provide the services at agreed-upon service levels - Safeguards the services' security and privacy	- Serves as an intermediary between : - Cloud providers and customers, facilitating connectivity and transport of cloud services - Network and telecommunication carriers handle the distribution of cloud services, - while a transport agent is a company that arranges for the physical delivery of storage devices like high-capacity hard drives - Customers can access the cloud through the network, telecommunication, and other access equipment provided by cloud carriers	- Assess the cloud provider's services in terms of performance: - Service level agreement compliance - Privacy Implications - Security controls - The entity that can perform objective examinations of cloud services in conformity with IT standards	- Cloud broker provides the following three types of services Service intermediation - Cloud broker improves services by refining some specific characteristics and providing value-added features to cloud consumers Service aggregation - Merges numerous facilities into one or more new services by providing data integration and ensuring secure data migration between the cloud consumer and the cloud provider Service arbitrage: - Unlike with service aggregation, the services being integrated or aggregated are not fixed - Service arbitrage Broker has the freedom to integrate services from multiple cloud providers

Load Balancing

- Virtualization process in which the **workload is spread** out across multiple nodes, clusters of computers, or other computing resources to accomplish peak resource utilization, maximize throughput, minimize job response time, and avoid system overload

Load Balancing

- If computing resources are efficiently utilized, the system's overall performance in speed will automatically improve, resulting in better quality, reduction in energy consumption and carbon emission, and ultimately lower costs

Load Balancing Objectives

- Improve the overall system performance
- Construct a fault-tolerant system
- Increase computing resource availability
- Maintain system consistency and stability
- Increase user satisfaction

Load Balancing Classification

- The System load
 - How the workload is evenly distributed among computing resources
- The System topology
 - How the data/information is utilized for effecting load balancing

Load Balancing Classification

How the workload is evenly distributed
among computing resources

Load Balancing Algorithms

How the data/information is utilized
for effecting load balancing

Based Upon System Load

Centralized

Distributed

Mixed

Based Upon System Topology

Static

Dynamic

Adaptive

Mobile Cloud Computing

Mobile Cloud Computing (MCC)

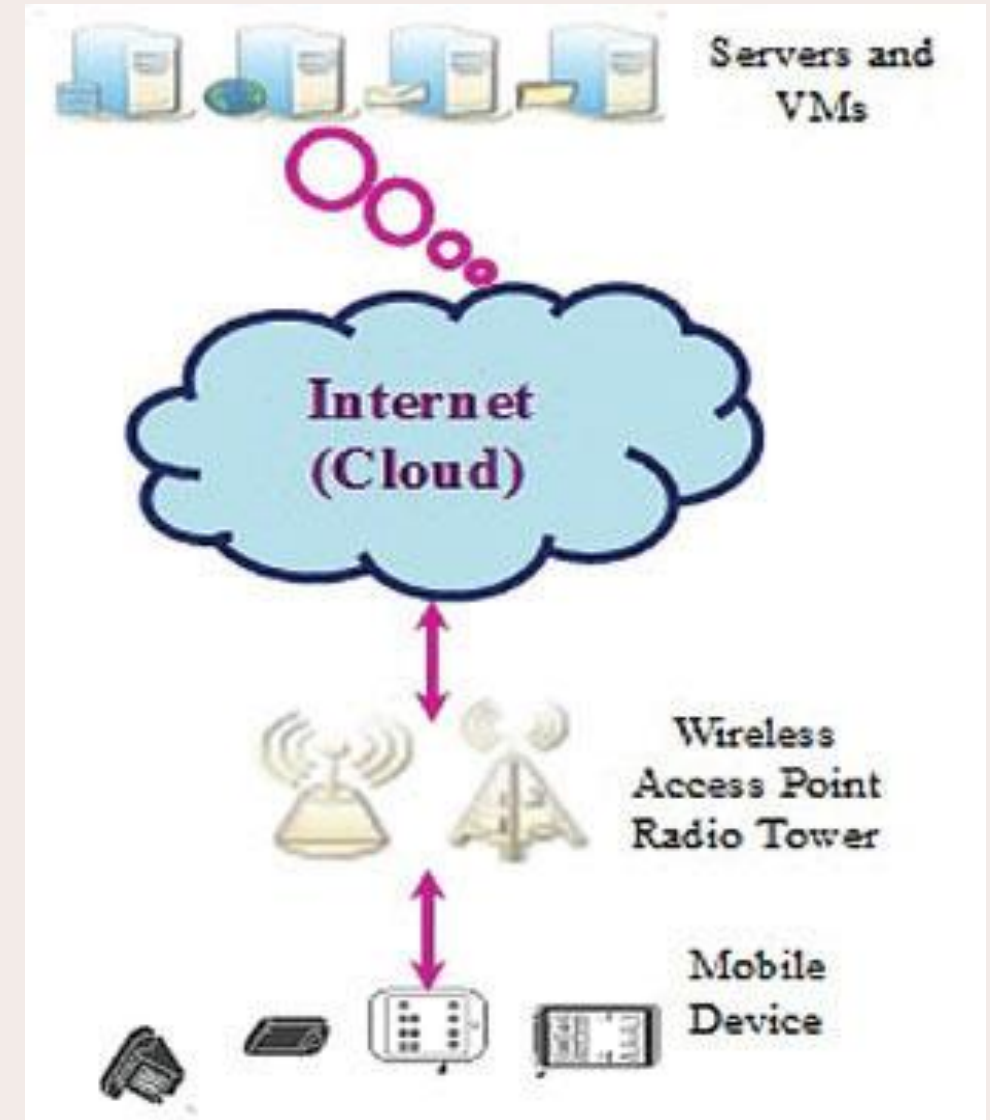
- The fundamental technology underlying cloud computing is **centralizing computing facilities** and **specific applications provided as a utility to end users**
- **Mobile Cloud Computing** (MCC) is the incorporation of **cloud computing** into **mobile device applications**
- **MCC** systems are **virtualized**, and **resource capacity** is distributed over a network of **interconnected computers** rather than in **traditional local computers or servers**

Mobile Cloud Computing

- Applications based on **mobile cloud computing** have been created and provided to users on Android, Apple, and Microsoft platforms encompassing mobile emails, navigation systems, social media apps, games, knowledge content, and productivity tools

Mobile Cloud Computing

- **Mobile computing** is comprised of three key concepts:
 - Hardware
 - Software
 - Communication



Mobile Cloud Computing

Hardware

- Can be grouped into several categories of **portable devices** and their **mobile components**

Software

- Consists of **mobile applications** installed on these devices, such as the **mobile browser**, **anti-virus software**, **diagnostics**, and **games**

Communication

- Includes the **infrastructure** of mobile networks, **access point protocols**, and **data packet delivery**

Mobile Cloud Computing Features

- Mobility
- Diversity of network conditions
- Frequent disconnection and consistency
- Dis-symmetrical network communication
- Low reliability

Mobile Cloud Computing **Features**

Mobility	Diversity of network conditions	Frequent disconnection and consistency	Dis-symmetrical network communication	Low reliability
▪ Each node within the mobile computing network can establish a connection with others ▪ Even fixed nodes in wired networks can connect through a Mobile Support Station (MSS)	• Networks utilized by mobile nodes can be a wired network with high bandwidth, or a Wireless Wide Area Network (WWAN) with low bandwidth	• Mobile nodes do not always maintain their connection. • They disconnect from the wireless networks passively or actively based on bad network conditions, limited battery power	▪ Servers, access points, and other MSS permit strong send/receive signals ▪ However, such ability in mobile nodes is comparatively weak. ▪ Consequently, the communication bandwidth and overhead between downlink and uplink are incongruent	Mobile data signals are more vulnerable to interference

Advantages of Mobile Cloud Computing

- Improving data storage capacity and processing power
- Improving reliability
- Multi-tenancy
- Ease of integration

Advantages of Mobile Cloud Computing

Advantages	Description
Improving data storage capacity and processing power	Mobile Cloud Computing - developed to enable mobile users to store/access cloud data
Improving reliability	Storing data on multiple computers in the cloud improves reliability On-demand provisioning of resources and scalability allow applications to be accessed without reserving computing resources or paying for additional services
Multi-tenancy	Service providers can pool resources to support an assortment of applications with a multiplicity of users
Ease of integration	Services from diverse service providers can be assimilated through the cloud

Advantages of Mobile Cloud Computing



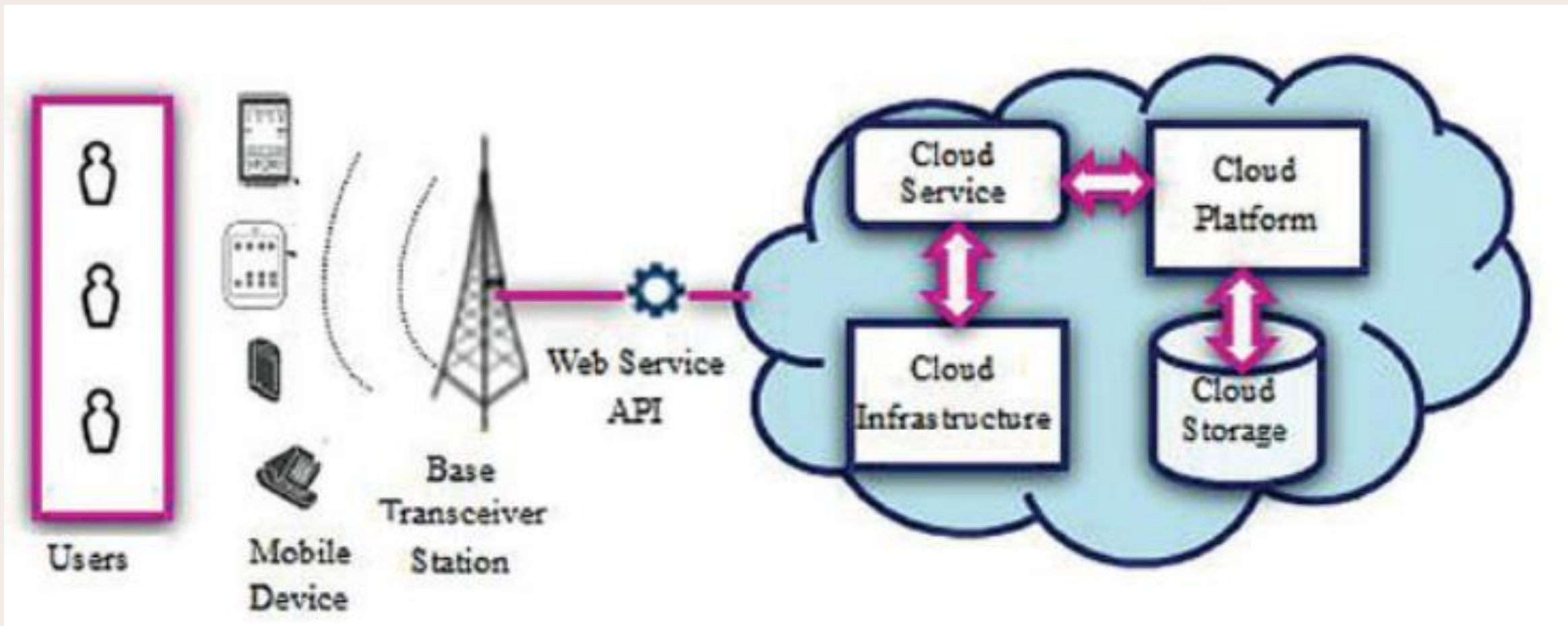
Disadvantages of Mobile Cloud Computing

- Due to the **wireless environment** and **numerous mobile nodes**, mobile computing networks may **face various complications** than a **conventional wired network**

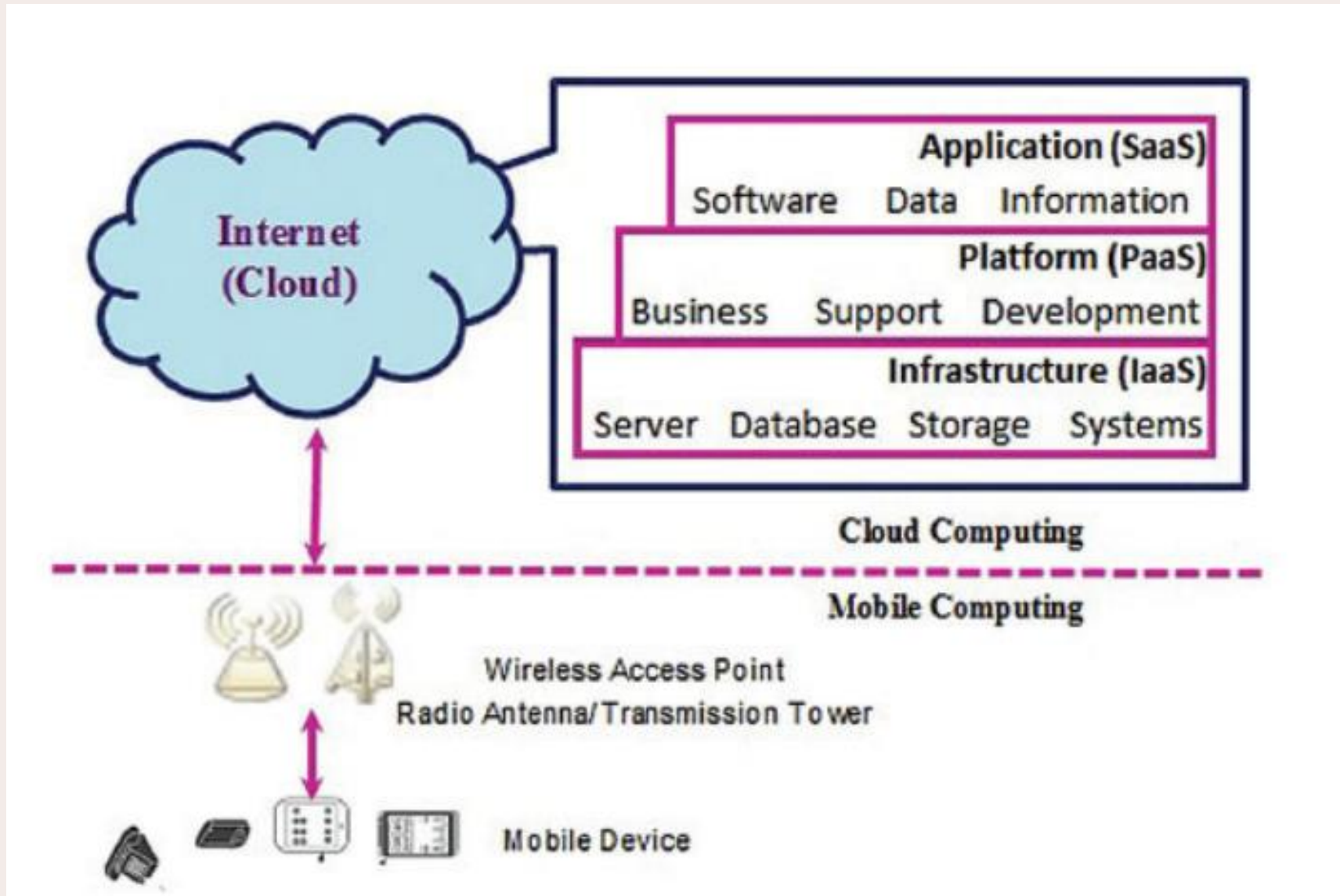
Disadvantages of Mobile Cloud Computing

- Signal disturbance from landforms
 - Weather
 - Buildings
 - Security
- Handoff delay, limited power, low computing ability

Mobile Cloud Computing Architecture



Mobile Cloud Computing Architecture



Case Study

Case Study: Oracle Cloud Management

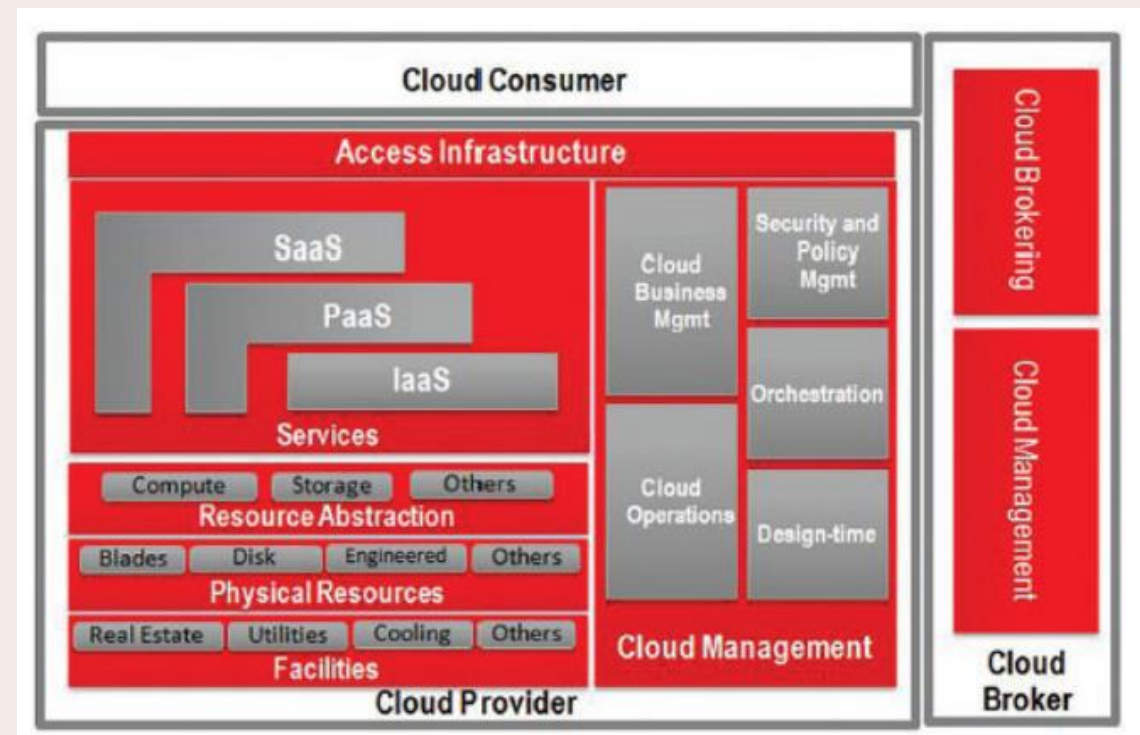
- **Oracle management cloud** is a suite of management services that eliminates the human effort associated with conventional solutions for:
 - Monitoring
 - Managing
 - Safeguarding applications and infrastructure

Oracle Cloud



Conceptual View of Cloud Computing

- The Oracle Cloud Conceptual View
 - Illustrates three key cloud perspectives:
 - Cloud Provider
 - Cloud Consumer
 - Cloud Broker



Logical View of Cloud Computing

- The access layer splits into two important sets of functionality:
 - **Interfaces**
 - The cloud needs a variety of **user interfaces** to enable access to the **underlying services** as well as its **management capabilities**
 - **Network components**

Oracle Cloud Mapping to the Logical View

Oracle provides one of the most **comprehensive** and **integrated** cloud software and hardware products to facilitate **public, private, and hybrid** clouds

For **private PaaS** clouds, Oracle offers an extensive collection of industry applications that run on **standards-based, shared services** platforms

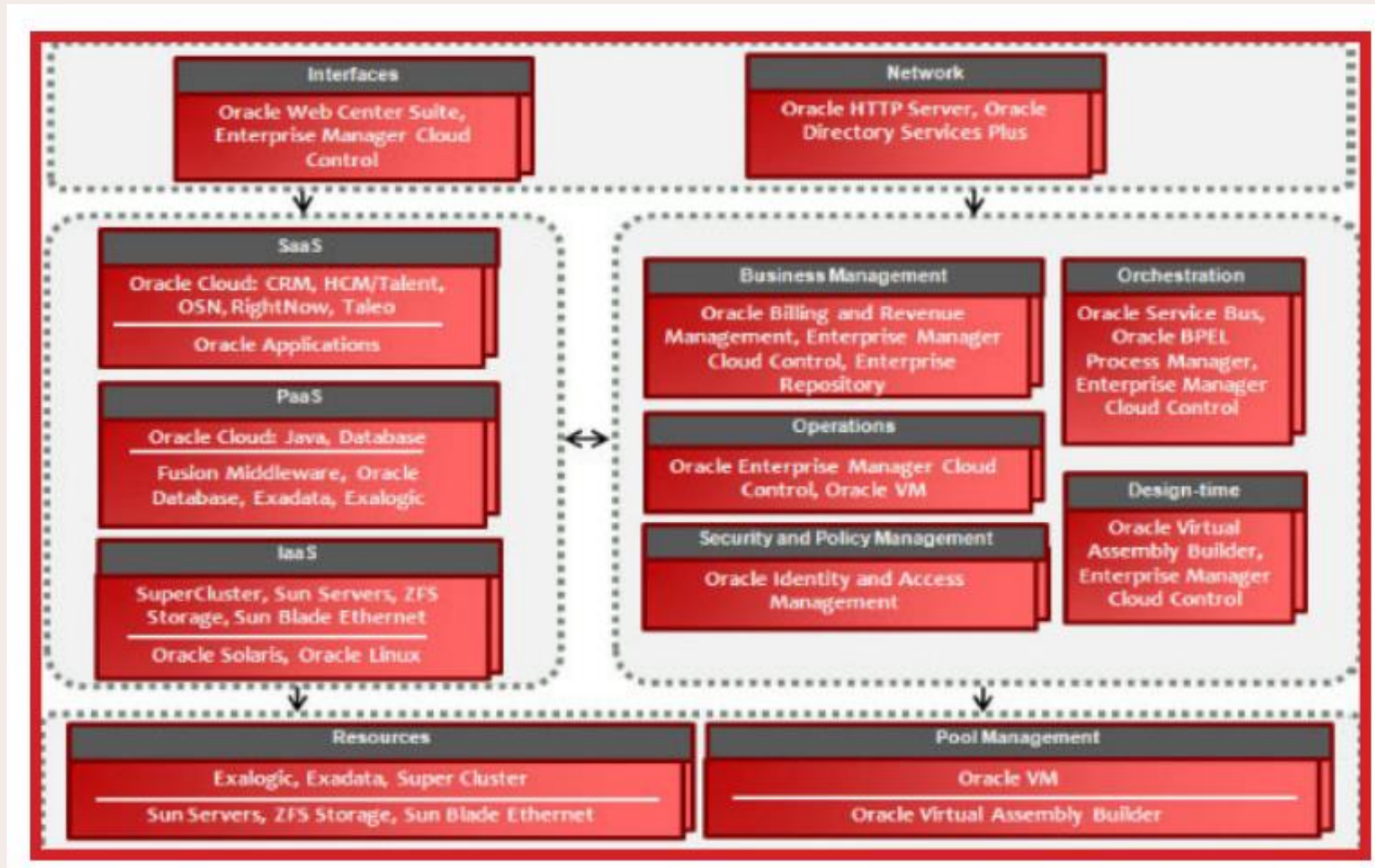
Engineered systems such as Oracle Exadata Database Machine, and Oracle Exalogic Elastic Cloud, provide high-performance platforms for building platform services

Oracle Cloud Mapping to the Logical View

For **private IaaS** clouds, Oracle offers **leading computing hardware resources**.

Customers can also choose to run **new generation Oracle applications** through third-party public clouds using **on-premise** or **off-premise private models**

Oracle Cloud Mapping to the Logical View



Summary

- Cloud computing is one of the technologies with great benefits:
 - making storage and access of data easier and more efficient
- The **objective of the CCLC** is to negotiate the ever-changing nature of the cloud environment, expedite service speed, increase flexibility, and promptly satisfy the needs of individuals and organizations

Summary

- There are four stages/phases within the cloud life cycle:
 - Architect
 - Engage
 - Operate and
 - Fresh

Summary

The NIST cloud computing reference architecture defines five major actors:

- Cloud Consumer
- Cloud Provider
- Cloud Carrier
- Cloud Auditor, and Cloud Broker

Summary

Load balancing is a virtualization process in which the workload is spread out across multiple nodes, clusters of computers, or other computing resources to accomplish peak resource utilization, maximize throughput, minimize job response time, and avoid system overload

Summary

The **load-balancing algorithm** can be classified into **two major categories**:

- System load
- System topology

Summary

Mobile Cloud Computing:

The incorporation of cloud computing into mobile device applications

Summary

Mobile computing is comprised of three key concepts:

Hardware

Software

Communication

Summary

Mobile cloud computing architecture offers a model that links the benefits of both **mobile technology** and **cloud computing**

Summary

Oracle management cloud is a suite of management services that **eliminates** the human effort associated with **conventional solutions** for monitoring, managing and safeguarding applications and infrastructure

Summary

Cloud consumer/client: An individual or organization that retains a professional relationship with and uses service from cloud service providers

Cloud Provider: An entity responsible for making cloud services available to clients

Summary

Cloud Carrier: The intermediary that provides connectivity and transference of cloud services between cloud providers and cloud consumers

Cloud Broker: An entity that manages the use, performance, and delivery of cloud services, and negotiates relationships between cloud providers and cloud consumers

Cloud Auditor: A neutral party that can conduct objective, independent assessments of cloud services, information system operations, performance, and security of the cloud implementation

*What have we learned
today?*

Q & A

Thank you



Next Lesson: Cloud Computing Virtualization Technology